

Rawja Alam

📍 Dhaka, Bangladesh ✉ rawja.alam@gmail.com 📄 Google Scholar 🏠 ORCID

Academic Credentials

Military Institute of Science and Technology

Feb 2020 – April 2024

Bachelor of Science in Civil Engineering (Major: Water, Minor: Structure)

- GPA: 3.47/4.0
- **Coursework:** River Engineering, Hydraulic Structures, Hydraulic Structures Design Sessional, GIS and Remote sensing, Engineering Hydrology and Irrigation Engineering, Design of Water Supply, Sanitation, Fundamentals of Transportation Engineering, Highway Materials, Pavement Design, Railway Engineering, Traffic Engineering, Finite Element Method, Design of Steel-Concrete Composite Structure

Standardized Test

International English Language Testing System (IELTS Academic)

12 Oct, 2024

Overall Band Score: 7.5

L: 8.0 R: 8.0 W: 6.5 S: 7.0

Graduate Record Examination (GRE)

21 Nov, 2024

Total: 302 Quantitative: 157 Verbal: 145 Analytical Writing: 3.5

Work Experience

Research Assistant

Dhaka, Bangladesh

Bangladesh University of Engineering and Technology (BUET)

Nov 2025 - Present

- Working on Web-based Agricultural Information System for Bangladesh (WebAIS) Project
- Developing a DDIEI-inspired, grid-based water and salt balance model in MATLAB for Polder 30
- Integrating hydro-climatic, land-use, cropping, flow-routing, and salinity data for daily soil salinity prediction

Internship

Dhaka, Bangladesh

Ministry of Health and Family Welfare (Health Engineering Department)

February 2023

- Monitored design details, site construction process, and cost estimation of work
- Observed the specific methods in sheet pile construction, slab and shutter design, and site safety measures and highlighted disparities between hospital and general construction processes
- Prepared presentations and documents for the project and BOQ

Publications

Assessing Riverbank Transformations of the Major Rivers in Bangladesh: A Remote Sensing and GIS based Study of Erosion and Accretion Patterns from 1988-2023

April 2025

Quazi Afra Nawar, **Rawja Alam**, Jumana Akhter, G. M. Jahid Hasan

DOI: [10.1080/01490419.2025.2494695](https://doi.org/10.1080/01490419.2025.2494695) [🔗](#)

Redefining Value Creation: Assessing the Economic Impact of Digital Twin Across Industries (Accepted)

IEOM 2024 World

Congress, Detroit,

Md Tarique Hasan Khan, Saki Rezwana, **Rawja Alam**

Michigan, October, 2024

Multi-Sectoral Impact Assessment and Flood Risk Zoning of Eastern Bangladesh for the August 2024 Flood (Accepted)

ICCEI 2025

Naimul Hasan, Nusaiba Nueri Nasir, Mairukh Wardi, **Rawja Alam**, Chandan Kumar Sarker, Rakibuzzaman Shafi Siam, Kh. Al Awual Arabi, Aditya Azim

Research Work

ASSESSING THE BANKLINE DYNAMICS OF THE MAJOR RIVERS OF BANGLADESH USING GIS AND DSAS TECHNIQUES (Thesis)

This study investigates the dynamics of riverbank erosion and accretion in the major rivers of Bangladesh using remote sensing and GIS techniques. Landsat satellite imagery was collected and processed, and NDVI (Normalized Difference Vegetation Index) was applied to evaluate vegetation cover and environmental conditions along riverbanks. River boundaries were digitized to generate GIS-based spatial datasets for hydrological analysis. The Digital Shoreline Analysis System (DSAS) was used to calculate erosion and accretion rates across different time scales, enabling the quantification and assessment of riverbank changes under both short-term and long-term conditions.

MOTORCYCLE OWNERSHIP AND USAGE PATTERNS AMONG INCOME GROUPS IN DHAKA: IMPLICATIONS FOR PUBLIC TRANSPORTATION SYSTEMS (Ongoing)

This ongoing study explores the distribution and utilization of motorcycles across different income groups in Dhaka. The project aims to understand how rising motorcycle ownership influences travel behavior and affects the performance and planning of public transportation systems.

REVIEW ON EARTH-CORE ROCKFILL DAMS (Ongoing)

I am co-authoring a review paper focusing on the structural stability and environmental challenges associated with earth-core rockfill dams. My contributions include collecting and organizing published research data, conducting a scientometric analysis to assess global research trends, and critically evaluating design structures, performance limitations, and resilience under hydrological and seismic stresses.

GLOBAL DAM MAPPING USING GIS (Ongoing)

In this ongoing project, I am developing an interactive GIS-based global dam database. This work involves collecting dam location and attribute data from international sources and satellite imagery, processing spatial datasets, and creating visualizations to highlight dam distribution and characteristics.

Technical Strengths

Software Skills: AutoCAD, SAP2000, ETABS, HECRAS, ArcGIS Pro, ArcGIS Desktop, QGIS

Programming Languages: C/C++, Python, MATLAB

Others: Latex, Unreal Engine, Adobe Illustrator, Microsoft Excel

Achievement and Extra Curricular Activities

University Merit Scholarship in MIST (Level-2)

- In recognition of academic excellence in Undergraduate studies

Elected as Class Representative, Department of Civil Engineering, MIST

- By votes of students and recommendation of faculty members (2020)

Certifications

Reviewer Certificate, *Catena*

March-November 2025

- Reviewed 7 manuscripts on climate-driven vegetation changes and sediment-discharge modeling.

Global Environmental Management, *Coursera* *Technical University of Denmark (DTU)*

June 27, 2020

Introduction to Engineering Mechanics, *Coursera* *Georgia Institute of Technology*

June 21, 2020

C for Everyone, Part 1: Programming Fundamentals, *Coursera* *University of California, Santa Cruz*

June 13, 2020

References

Dr. G M Jahid Hasan

Professor

Department of Civil Engineering

Military Institute of Science and Technology (MIST)

Email: jahid@ce.mist.ac.bd

Dr. Abul Fazal M. Saleh

Professor

Department of Environmental, Water Resources and Coastal Engineering

Military Institute of Science and Technology (MIST)

Email: saleh@iwfm.buet.ac.bd

Dr. Saki Rezwana

Assistant Professor

Department of Mechanical, Environmental and Civil Engineering

Tarleton State University

Email: SREZWANA@tarleton.edu